Sourav Minhas

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EDUCATION

Carleton University, Department of Computer Science Sep 2022 – Apr 2027
Honours Bachelor of Computer Science: Artificial Intelligence & Machine Learning - Co-op *Ottawa, Ontario*

- **Awards & Achievements:** Dean's Honour List, General In-Course Scholarship, VRC Robotics Design & Build Award
- **Minor, Standing & GPA:** Mathematics, Fourth Year, 10.0/12.0 (A-)

TECHNICAL SKILLS

Languages: Python, Java, Go, C++, C, JavaScript, TypeScript, SQL, Bash, Scheme, Prolog
Libraries: PyTorch, TensorFlow, pandas, scikit-learn, Matplotlib, Hugging Face Transformers, OpenCV
Frameworks & Tools: React, Node.js, Spring Boot, Playwright, JUnit, Git, GitHub, GitLab, Postman, SQLite
DevOps & Cloud: Docker, Podman, Kubernetes (OpenShift), Helm, Ansible, Jenkins, OpenStack, AWS, GCP

WORK EXPERIENCE

RBC May 2026 – Present
Quality Engineer Intern *Toronto, Ontario*

- Automated regression tests using **Playwright** for POS reporting workflows within the Merchant Services Workbench
- Contributed to test development in **TypeScript** by debugging failing test cases and validating web application behaviour
- Supported release validation by executing automated regression tests daily across pre-prod and production environments

Nokia Jan 2026 – Apr 2026
Automation Engineer Co-op *Ottawa, Ontario*

- Engineered the full software development lifecycle of a **Go**-based CLI product feature within the **NEDR** platform, from architecture discussions and technical design to implementation, documentation, and reviews with all product owners
- Improved unit testing workflows for containerized services using **Java**, **JUnit**, and **Spring Boot** by debugging failing test cases, validating system behaviour, and resolving security and automation-related issues across many environments
- Supported **CI/CD** workflows using **Docker**, **Kubernetes**, **Jenkins**, **Helm**, and **Ansible**, supporting cloud and baremetal environments while collaborating within an Agile/Scrum development team through sprint planning and daily standups

Software Support Co-op Sep 2025 – Dec 2025

- Resolved **Jira** tickets for the **NSP** platform involving network and device management, API calls via **Postman**, workflow issues, DR setups, database restores in **SQL/PostgreSQL**, and IP and MDM configurations across deployments
- Deployed **OpenStack** VMs hosting cloud-native applications, using **Kubernetes**, **Podman**, and **Linux** to troubleshoot systems, monitor services via **Grafana/Prometheus**, and validate endpoints, enhancing product reliability for customers
- Debugged and maintained platform code, implementing fixes in **Python** and **Bash**, documenting changes, and integrating updates into the NSP codebase via **GitLab** and **Jenkins**, improving system reliability and long-term maintainability

Pepperdata Jun 2025 – Aug 2025
Software Engineer Intern *Toronto, Ontario*

- Developed scalable **PyTorch** workloads on **AWS** and **GCP** with **Kubernetes**, building reproducible training and benchmarking pipelines with **Docker** and **Jenkins** to support distributed ML experimentation and infrastructure automation
- Implemented internal tooling to benchmark GPU performance for machine learning workloads, running fine-tuning and inference experiments to collect utilization metrics, compare different configurations, and analyze multiple workloads
- Built a tool that transcribed **200+** videos using **Whisper** and leveraging **Vertex AI** with **GKE** and **EKS** to preprocess and fine-tune an **LLM**-based **QA model** that enabled employees to review design discussions quickly and effectively

PROJECTS

PiVision *QNX | Raspberry Pi 5 | Python | OpenCV*

- Built a distributed vision processing system with a **QNX**-based **Raspberry Pi 5** capturing video via **Camera Module 3**
- Implemented Windows server pipeline in **Python** using **OpenCV** to process videos uploaded over a local device network
- Evaluated detection results using **MediaPipe** confidence scores, observing consistent performance above an **80%** mark